Marcus Vance

Staff Machine Learning Infrastructure Engineer

Specialist in Large-Scale GPU Serving, CUDA Kernel Optimization & vLLM

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PROFILE

Staff ML infrastructure specialist with 9+ years architecting high-throughput GPU serving clusters, distributed model parallel training, and low-latency LLM inference pipelines. Deep expertise in CUDA kernel acceleration, vLLM continuous batching, FlashAttention, and high-dimensional vector search.

EXPERIENCE

Staff ML Infrastructure Engineer 04/2022 - Present

HyperScale AI - San Francisco

Architected GPU serving infrastructure for 70B+ parameter generative LLMs across 500+ H100 GPUs. Integrated vLLM PagedAttention and custom Triton CUDA kernels, cutting time-to-first-token by 55% while doubling sustained token throughput.

Stack: PyTorch, CUDA, vLLM, TensorRT-LLM, Triton, Slurm, NCCL, RDMA

Senior ML Platform Engineer 02/2018 - 03/2022

NeuralStack - Palo Alto

Built real-time dense vector embedding search platform processing 40,000 queries per second over 500M document vectors using HNSW approximate nearest neighbors and Libsepal indexing.

Stack: Python, C++, Vector DBs, HNSW, FAISS, PyTorch, Kubernetes

Machine Learning Engineer 07/2015 - 01/2018

VisionCore - San Jose

Trained computer vision convolutional models and optimized TensorRT inference pipelines on edge hardware for autonomous drones.

Stack: C++, CUDA, OpenCV, Python, TensorFlow, ONNX

Research Assistant 09/2013 - 06/2015

Stanford AI Lab - Stanford

Co-authored papers on distributed stochastic gradient descent algorithms and asynchronous model parallel parameter servers.

Stack: Python, C++, CUDA, MPI, OpenMP

CORE SKILLS

Deep Learning & LLMs

[PyTorch] [vLLM] [PagedAttention] [TensorRT-LLM] [Triton Server] [Transformers] [FlashAttention]

GPU Systems & Parallelism

[CUDA] [C++] [Triton Lang] [NCCL] [Slurm] [Megatron-LM] [FSDP] [DeepSpeed ZeRO-3]

Vector & Retrieval Search

[Dense Embeddings] [HNSW] [FAISS] [Libsepal] [ANN Search] [Quantization FP8/INT4]

Cloud & Infrastructure

[Kubernetes] [Docker] [AWS EC2/EKS] [Prometheus] [Linux Kernel] [WandB]

SYSTEMS ARCHITECTURE & PROJECTS

paged-kernel-ext

High-performance custom CUDA FlashDecoding kernel for long-context KV cache management.

vector-mesh

Low-latency distributed vector indexing daemon in C++ handling billions of embedding vectors.

EDUCATION

M.Sc. Computer Science (AI)

Stanford University

LANGUAGES

English	Native
French	Working